Confidence Intervals & Hypothesis Testing

Econometrics TA session

27.10.26

1. Warm-up: conditional means

Conditional mean: the average *inside a group*

Idea. The overall average hides differences between groups. The conditional mean answers: *"what is the average of Y among those with $X = x$?"*

Tiny example

10 students. 6 attended the review session ($X = 1$), 4 did not ($X = 0$). Mean grade: 70 for attendees, 55 for the others.

$$E(Y \mid X = 1) = 70, \qquad E(Y \mid X = 0) = 55$$

- In a sample: just the average of the observations in that group.

Conditional mean from probabilities (discrete case)

When we have a joint probability table instead of raw data:

$$\Pr(Y = y \mid X = x) = \frac{\Pr(Y = y,\ X = x)}{\Pr(X = x)}$$

$$E(Y \mid X = x) = \sum_j y_j \Pr(Y = y_j \mid X = x)$$

Three-step recipe

1. **Pick the row** (or column) of the group $X = x$.
2. **Divide by its total** $\Rightarrow$ conditional probabilities (they now sum to 1).
3. **Weighted average** of the values y_j using those probabilities.

Same as the ordinary mean, but we first "zoom in" on one group and renormalise.

Law of Iterated Expectations: from groups back to the whole

Idea. The overall mean is the **weighted average of the group means**, where each group is weighted by *how large it is*.

Continuing the tiny example

$$E(Y) = \underbrace{\tfrac{6}{10}}_{\text{share}} \times \underbrace{70}_{\text{mean}} + \underbrace{\tfrac{4}{10}}_{\text{share}} \times \underbrace{55}_{\text{mean}} = 42 + 22 = 64$$

$$E(Y) = \sum_i E(Y \mid X = x_i)\ \Pr(X = x_i) \qquad \text{in a sample: } \bar{y} = \sum_g \frac{n_g}{n}\, \bar{y}_g$$

Exercise 1 · Conditional mean from a joint table

A café records, for each customer, the **number of coffees** Y and the **time of the visit** X. Joint probabilities:

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
Morning ($X=M$)	0.06	0.18	0.24	0.12
Afternoon ($X=A$)	0.12	0.12	0.10	0.06

1. Marginal distribution of X (add up each row).
2. Conditional distribution of $Y|X=M$ and $Y|X=A$ *(hint: divide rows by their total).*
3. $E(Y \mid X=M)$ and $E(Y \mid X=A)$.
4. $E(Y)$ via the LIE; check against the marginal of Y.

Solution · Exercise 1

1. $\Pr(M) = 0.06 + 0.18 + 0.24 + 0.12 = \mathbf{0.60}$; $\quad \Pr(A) = 0.12 + 0.12 + 0.10 + 0.06 = \mathbf{0.40}$.

2. Divide each row by its total:

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
$\Pr(Y \mid M)$	0.10	0.30	0.40	0.20
$\Pr(Y \mid A)$	0.30	0.30	0.25	0.15

3. $E(Y \mid M) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = \mathbf{1.70}$;
$E(Y \mid A) = 0(0.3) + 1(0.3) + 2(0.25) + 3(0.15) = \mathbf{1.25}$.

4. $E(Y) = E(Y \mid M)\Pr(M) + E(Y \mid A)\Pr(A) = 1.70(0.60) + 1.25(0.40) = \mathbf{1.52}$.
Most customers come in the morning, so $E(Y)$ lies closer to the morning mean.

Exercise 2 · LIE with group means and sample sizes

A survey of 300 students reports average weekly study hours by cohort:

Cohort	n_g	$\bar{y}_g$ (hours)
1st-year Bachelor	100	14
2nd-year Bachelor	150	18
Master	50	26

Compute the average study time of the whole sample, **without** the raw data. Then compute the **unweighted** average of the three group means: why is it different?

Hint: the weights are the group shares n_g/n.

Solution · Exercise 2

Weights are $n_g/n = 1/3,\ 1/2,\ 1/6$:

$$\bar{y} = \tfrac{100}{300}(14) + \tfrac{150}{300}(18) + \tfrac{50}{300}(26) = 4.667 + 9 + 4.333 = \mathbf{18.0} \text{ hours}$$

The simple average of the group means, $(14 + 18 + 26)/3 = 19.33$, is **wrong**: it treats the small Master group as if it were as large as the others. The LIE needs the **weights**.

The big picture: why are we doing this?

The situation. There is a population with an unknown mean μ (e.g. the average commuting time of *all* workers). We cannot measure everybody, so we draw a **sample** and compute $\bar{y}$.

The problem. A different sample would give a different $\bar{y}$. So $\bar{y}$ is only a *guess* of μ.

Two questions that statistics answers

1. **Confidence interval:** *"Given my sample, which values of μ are plausible?"* $\rightarrow$ a range instead of a single number.
2. **Hypothesis test:** *"Is a specific claim about μ (e.g. $\mu = 30$) compatible with my sample, or is it too surprising?"*

2. Sampling distribution

Why does $\bar{y}$ vary? The sampling distribution

Thought experiment. Draw a sample of n workers, compute $\bar{y}$. Throw it back, draw a *new* sample, compute a new $\bar{y}$. Repeat many times.

- Each repetition gives a **different** $\bar{y}$. So **before** we draw it, $\bar{y}$ is a *random variable*.
- The distribution of all these $\bar{y}$ values is the **sampling distribution** of $\bar{y}$.
- Our actual sample is **one draw** from that distribution. Everything in this session asks: *how far can one draw be from μ?*

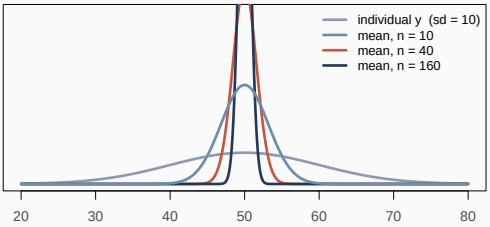
Estimator vs estimate

Estimator = the recipe ("take the average"), a random variable. **Estimate** = the number the recipe gives for *your* sample.

What we know about the sampling distribution of $\bar{y}$

For an i.i.d. sample with population mean μ and variance σ^2:

$$E(\bar{y}) = \mu \quad \text{(on average right: unbiased)} \qquad \mathrm{V}(\bar{y}) = \frac{\sigma^2}{n}$$



- **Averaging cancels noise:** high and low observations offset each other.
- Spread of $\bar{y}$ is $\sigma/\sqrt{n}$: $n = 10, 40, 160$ gives 3.2, 1.6, 0.8 (with $\sigma = 10$).

Figure 1: Individuals are spread out; sample means are concentrated, more so as n grows.

Estimating the unknown: s^2 and the standard error

We almost never know σ. We replace it by the sample version:

$$s^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2, \qquad \hat{V}(\bar{y}) = \frac{s^2_{n-1}}{n}$$

- **Why $n-1$?** The deviations are measured from $\bar{y}$, which was itself fitted to *these* data, so they are slightly too small. Dividing by $n-1$ ("degrees of freedom": one is used up by $\bar{y}$) corrects it exactly on average.

Exercise 3: Sample variance and standard error

A sample of $n = 26$ daily sales figures has $\sum_{i=1}^{26}(y_i - \bar{y})^2 = 1000$.

1. What is the sample variance s_{n-1}^2?
2. What is the standard error of $\bar{y}$?
3. Which of these is a common wrong answer, and what mistake produces it?
   (a) 1.240 (b) 1.538 (c) 6.325 (d) 38.46 (e) 40

Hint: divide by $n - 1$ first, then divide by n, then take the square root.

Solution Exercise 3

1. $s^2_{n-1} = \dfrac{1000}{n-1} = \dfrac{1000}{25} = \mathbf{40}$ (divide by $n-1$, not n).

2. $\mathrm{SE}(\bar{y}) = \sqrt{\dfrac{s^2_{n-1}}{n}} = \sqrt{\dfrac{40}{26}} = \sqrt{1.538} = \mathbf{1.240} \rightarrow$ **(a)**

3. The traps:

   - (b) $1.538 = s^2/n$: forgot the square root
   - (c) $6.325 = s$: forgot to divide by $\sqrt{n}$ (that is the sd, not the SE)
   - (d) $38.46 = 1000/26$: divided by n instead of $n-1$ (and no root)
   - (e) 40: that is s^2, not the SE

3. Confidence intervals

From a point estimate to an interval

Idea. $\bar{y}$ is a single guess. We know it is typically off by about one standard error, so we report a range around it:

$$\boxed{\text{estimate} \ \pm \ (\text{multiplier}) \times (\text{standard error})}$$

- **Standard error** = how noisy our estimate is.
- **Multiplier** (critical value) = how *confident* we want to be. More confidence $\Rightarrow$ bigger multiplier $\Rightarrow$ wider interval.

Why is the multiplier 1.96 for 95%?

If $\bar{y}$ is normal around μ, then **95% of the time** it falls within ± 1.96 standard errors of μ (figure on the next slide). So μ is within ± 1.96 SE of $\bar{y}$ in 95% of samples.

Where 1.96 comes from

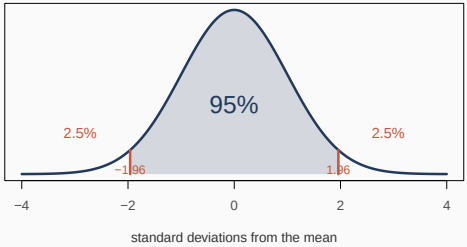


Figure 2: The central 95% of a standard normal lies within ± 1.96. Leaves 2.5% in each tail.

- Notation: confidence level $= 1 - \alpha$. For 95%, $\alpha = 0.05$, split equally between the two tails.
- The multiplier is $z_{1-\alpha/2}$: for 95% we need the **0.975 quantile** (leaving 2.5% in the upper tail).
- 90% $\rightarrow \pm 1.645$ 95% $\rightarrow \pm 1.960$ 99% $\rightarrow \pm 2.576$

Deriving the interval (known σ)

With $z = z_{1-\alpha/2}$, we start from what we know about the standardised mean, $\frac{\bar{y} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$, and rearrange to isolate μ:

$$\Pr\left(-z \leq \frac{\bar{y} - \mu}{\sigma/\sqrt{n}} \leq z\right) = 1 - \alpha \iff \Pr\left(\bar{y} - z\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{y} + z\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

The $(1 - \alpha)$ confidence interval

$$\bar{y} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Look at what is random in the second probability: **not** μ (a fixed number), but the endpoints, because they contain $\bar{y}$.

But we don't know σ: which distribution?

Standardising = "how many standard errors away from μ is $\bar{y}$?" Replacing σ by s changes the answer:

Situation	Statistic	Distribution
σ^2 **known**, Y normal	$\dfrac{\bar{y}-\mu}{\sigma/\sqrt{n}}$	$N(0,1)$ **exactly**
σ^2 **known**, Y any, n large	$\dfrac{\bar{y}-\mu}{\sigma/\sqrt{n}}$	$\approx N(0,1)$ (CLT)
σ^2 **unknown**, Y normal	$\dfrac{\bar{y}-\mu}{s_{n-1}/\sqrt{n}}$	t_{n-1} **exactly**
σ^2 **unknown**, Y any, n large	$\dfrac{\bar{y}-\mu}{s_{n-1}/\sqrt{n}}$	$\approx N(0,1)$ (CLT)

- **Why t in the third row?** *two* sources of uncertainty: $\bar{y}$ is random **and** s is random. The t distribution has fatter tails to account for the extra uncertainty.
- **Small n needs normal data**; large n is rescued by the CLT.

What does “95% confident” really mean?

- A CI is a **random interval**: its endpoints are statistics, so each sample gives a different one.
- It contains the unknown μ with the **preassigned probability** $1 - \alpha$. For 95%: about 95 intervals out of 100 contain μ.

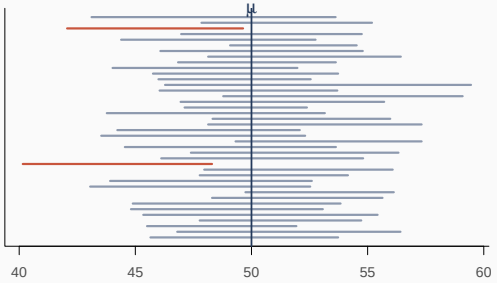


Figure 3: 40 samples, 95% CIs. Red = interval misses μ (about 1 in 20).

z or t?

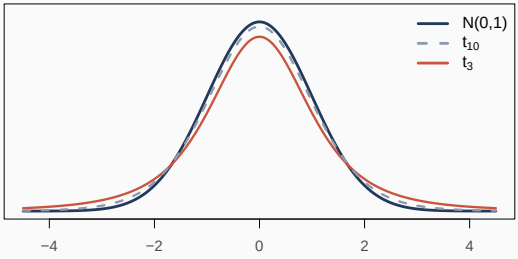


Figure 4: t has fatter tails (extra uncertainty from estimating σ); t_{10} is already close to $N(0,1)$.

- Fatter tails $\Rightarrow$ larger critical values: for 95%, $t_8 = 2.306$ vs $z = 1.96$.
- As $n-1$ grows, $t_{n-1} \to N(0,1)$: $t_{30} = 2.042$, $t_{120} = 1.980$, $t_\infty = 1.960$.
- With normal data, t is **never wrong**; for large n the two give practically the same interval, so we use z.

Critical values · z and t distributions

Standard normal z

One-tailed α	Two-tailed α	Critical value
0.10	0.20	1.282
0.05	0.10	1.645
0.025	0.05	1.960
0.01	0.02	2.326
0.005	0.01	2.576
0.001	0.002	3.090

Student's t

One-tailed α	0.10	0.05	0.025	0.01	0.005
Two-tailed α	0.20	0.10	0.05	0.02	0.01
df = 1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
8	1.397	1.860	2.306	2.896	3.355
10	1.372	1.812	2.228	2.764	3.169
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
25	1.316	1.708	2.060	2.485	2.787
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Confidence intervals use the **two-tailed** row (95% CI $\rightarrow$ two-tailed $\alpha = 0.05$). The one-tailed row is for one-sided tests (Section 4). As df $\rightarrow \infty$, t converges to z.

Decision guide: which formula?

1. Is σ **given** in the problem (not just s), and Y normal or n large? $\rightarrow$ use σ and z.
2. Otherwise we use s_{n-1}. Is n large (say > 30)? $\rightarrow$ use z (CLT).
3. Is n small? $\rightarrow$ the data must be **normal**, and we use t_{n-1}.

Then always

1. **Standard error:** $\sigma/\sqrt{n}$ or $\sqrt{s_{n-1}^2/n}$ (the root comes *after* dividing by n).
2. **Critical value:** two-sided $\Rightarrow$ quantile $1-\alpha/2$ (95% $\rightarrow$ 0.975, *not* 0.95).
3. **Margin** $E = c \times \mathrm{SE}$. Lower $= \bar{y} - E$, upper $= \bar{y} + E$, **width** $= 2E$.

Worked example · a CI step by step

Hourly wage of young workers. Sample: $n = 100$, $\bar{y} = 20$ EUR, $s = 5$ EUR. σ unknown, n large.

Step by step

1. **Case:** σ unknown, n large $\Rightarrow z$.
2. **SE:** $s/\sqrt{n} = 5/\sqrt{100} = 5/10 = 0.5$.
3. **Critical value, 95%:** 1.96. **Margin:** $1.96 \times 0.5 = 0.98$.
4. **CI:** $20 \pm 0.98 = [19.02,\ 20.98]$.

Reading it: values of μ between about 19 and 21 EUR/hour are consistent with our data.

What if we want 99%? Multiplier 2.576: $20 \pm 1.288 = [18.71,\ 21.29]$. Wider: more confidence, less precision.

Exercise 4: CI, unknown σ^2, large sample

The commuting time of workers in a city is studied. A random sample of $n = 144$ workers gives $\bar{y} = 38.5$ minutes and $s_{n-1}^2 = 324$.

1. Why can we use the normal approximation here, even without knowing the population distribution?
2. Compute the standard error.
3. Build a 95% confidence interval for the mean commuting time.
4. What is its width?

Hint: follow the four steps of the worked example. Careful: the variance is given, not the standard deviation.

Solution Exercise 4

1. $n = 144$ is large: by the CLT $\bar{y}$ is approximately normal, and replacing σ by s_{n-1} is harmless in large samples.
2. $\mathrm{SE}(\bar{y}) = \sqrt{324/144} = \sqrt{2.25} = \mathbf{1.5}$ min.
3. $z_{0.975} = 1.96$:

$$\bar{y} \pm 1.96 \times 1.5 = 38.5 \pm 2.94 = [\mathbf{35.56},\ \mathbf{41.44}]$$

4. Width $= 2 \times 2.94 = \mathbf{5.88}$ min.

Interpretation: 95% of intervals built this way contain the true mean commute, which here is somewhere around 36 to 41 minutes.

Exercise 5: CI with *known* σ^2

A machine fills rice bags. Filled weights are normally distributed with a **known** standard deviation $\sigma = 20$ g. A sample of $n = 25$ bags has mean $\bar{y} = 502$ g.

1. Compute a 90% confidence interval for the mean filled weight.
2. Compute a 99% confidence interval. How does the width change, and why?

Hint: σ is given, so no s and no t: the standard error is $\sigma/\sqrt{n}$.

Solution Exercise 5

Standard deviation of $\bar{y}$ (no estimation needed): $\sigma/\sqrt{n} = 20/5 = 4$ g.
Since Y is normal: Exact $N(0,1)$

1. 90%: $z_{0.95} = 1.645$: $\quad 502 \pm 1.645 \times 4 = 502 \pm 6.58 = [\mathbf{495.42},\ \mathbf{508.58}]$
2. 99%: $z_{0.995} = 2.576$: $\quad 502 \pm 2.576 \times 4 = 502 \pm 10.30 = [\mathbf{491.70},\ \mathbf{512.30}]$

   Width grows from 13.16 to 20.61: higher confidence $\Rightarrow$ larger critical value $\Rightarrow$ wider interval.

Exercise 6: CI in a small sample

The battery life (hours) of a new phone is assumed normally distributed. A lab tests $n = 9$ phones: $\bar{y} = 11.2$ and $s^2_{n-1} = 2.25$.

1. Which distribution and which critical value should you use for a 95% CI? Why not 1.96?
2. Build the 95% confidence interval.
3. Which of these critical values is right for **a 99% CI with** $n = 16$?
   (a) 2.576 (b) 2.602 (c) 2.947 (d) 2.131 (e) 3.286

Hint: three ingredients for a critical value: which distribution, which quantile $(1 - \alpha/2)$, how many degrees of freedom $(n - 1)$.

Solution Exercise 6

1. $n = 9$ is small and σ^2 unknown $\Rightarrow \dfrac{\bar{y} - \mu}{s/\sqrt{n}} \sim t_8$ (needs normal data).
   $t_{8;\,0.975} = \mathbf{2.306}$. Using 1.96 would give an interval that is **too narrow**: it would cover μ less than 95% of the time.

2. $\mathrm{SE} = \sqrt{2.25/9} = 0.5$. Margin $= 2.306 \times 0.5 = 1.153$:

$$11.2 \pm 1.153 = [\mathbf{10.05},\ \mathbf{12.35}]$$

   (With z it would wrongly be $[10.22,\ 12.18]$.)

3. Two-sided 99% $\Rightarrow 1 - \alpha/2 = 0.995$; $df = n - 1 = 15 \Rightarrow t_{15;0.995} = \mathbf{2.947} \rightarrow$ **(c)**.
   (a) is the normal quantile; (b) is the one-sided $t_{15;0.99}$; (d) is $t_{15;0.975}$ (the 95% value).

Margin of error and sample size

The margin of error of the interval is $E = z_{1-\alpha/2} \times \mathrm{SE}(\bar{y}) = z_{1-\alpha/2}\sqrt{\sigma^2/n}$, so for a fixed confidence level and population:

$$E \propto \frac{1}{\sqrt{n}}$$

- Precision grows with $\sqrt{n}$, not with n: to make E k **times smaller**, you need $n' = n \times k^2$.
- Intuition: a 3 times smaller margin needs 9 times more data.
- Writing the ratio: $E'/E = \sqrt{n/n'} = 1/k \;\Rightarrow\; n' = k^2 n$.

Exercise 7: Sample size

1. A survey of $n = 150$ observations gives a CI with margin of error $E = z_{1-\alpha/2}\,\mathrm{SE}(\bar{y})$. For the same population and confidence level, how large must the sample be for the margin of error to be $E/3$?
2. A poll with $n = 80$ has margin E. What sample size gives a margin $E/2$?
3. The sample is doubled. By what factor does the width of the CI shrink?

Hint: $E \propto 1/\sqrt{n}$, so a k-fold smaller E needs k^2 times more data.

Solution Exercise 7

1. $E'/E = \sqrt{n/n'} = 1/3 \Rightarrow n' = 9n = 150 \times 9 = \mathbf{1350}$. Classic wrong answers: 450 ($3n$) or 50 ($n/3$, which makes the interval *wider*).
2. $n' = 2^2 \times 80 = \mathbf{320}$.
3. width $\propto 1/\sqrt{n}$: doubling n shrinks the width by $1/\sqrt{2} \approx 0.71$, i.e. only about **29%** narrower.

4. Hypothesis testing

The logic of a test: a trial analogy

A court case

- Default: the defendant is **innocent** until proven guilty.
- We convict only if the evidence is **very unlikely under innocence**.
- Not guilty $\neq$ proven innocent: maybe just not enough evidence.

A statistical test

- Default: H_0 is true (nothing special, e.g. $\mu = \mu_0$).
- We reject H_0 only if the data would be **very unlikely if H_0 were true**.
- Not rejecting $\neq$ proving H_0: maybe just too little data.

The one question a test asks

"If H_0 were true, how surprising would my sample be?" Very surprising $\Rightarrow$ we doubt H_0.

Setting up a test of the mean

- **Null** $H_0: \ \mu = \mu_0$ (the claim being checked) vs an **alternative**: two-sided $H_A : \mu \neq \mu_0$, or one-sided $H_A : \mu > \mu_0$ / $H_A : \mu < \mu_0$.
- Fix the significance level α (usually 5%) **before** looking at the data: the false-alarm rate we tolerate.
- **Test statistic** = *signal-to-noise ratio*: how many standard errors is $\bar{y}$ away from the claimed μ_0?

$$t = \frac{\bar{y} - \mu_0}{\mathrm{SE}(\bar{y})} = \frac{\text{signal: distance from } \mu_0}{\text{noise: typical sampling error}}$$

- If H_0 is true, $t \sim N(0,1)$ (large n) or t_{n-1} (normal Y, small n): the **null distribution**.
- $|t|$ close to 0: consistent with H_0. $|t|$ large: hard to explain by chance alone.

Decision rule and p-value

- **p-value:** the probability, **assuming H_0 is true**, of a test statistic *more extreme* than the one we observed.
- **Significance level** α: a cutoff on the p-value, chosen *before* the test. If $p < \alpha$ we **reject** H_0: what we observe is too unlikely under the Null.
- **Equivalent rule (critical value):** reject if the observed statistic falls beyond the critical value.
  - two-sided $H_A : \mu \neq \mu_0$: reject if $|t| > c_{1-\alpha/2}$ (95%: 1.96).
  - one-sided H_A: "more extreme" means only the direction of H_A (right or left), and the critical value is $c_{1-\alpha}$ (e.g. 1.645).

p-value: how surprising is my sample?

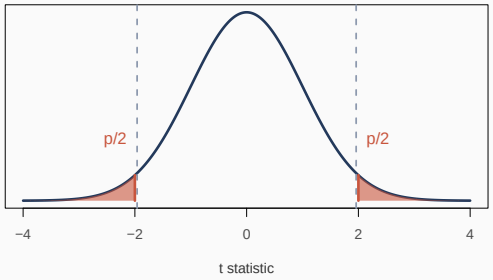


Figure 5: Two-sided test, $t = 2.0$: p-value = red area (both tails).

- **p-value** = probability, **assuming** H_0 **is true**, of getting a result *at least as extreme* as ours.
- Small p $\Rightarrow$ our data would be rare under H_0 $\Rightarrow$ doubt H_0.
- **Reject** $\iff$ $p < \alpha$ $\iff$ $|t| >$ **critical value.** Two routes, same decision.

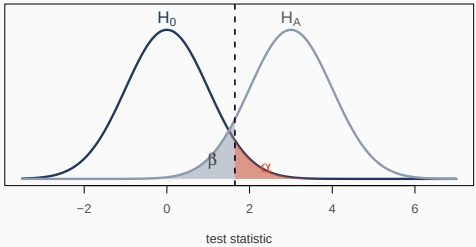
Worked example · a test step by step

Does the average student sleep **8 hours**? Sample: $n = 100$, $\bar{y} = 7.7$ h, $s = 3$ h.

Step by step

1. **Hypotheses:** $H_0 : \mu = 8$ vs $H_A : \mu \neq 8$. Choose $\alpha = 5\%$.
2. **SE:** $3/\sqrt{100} = 0.3$.
3. **Test statistic:** $t = \dfrac{7.7 - 8}{0.3} = -1.0$: the sample is only **one SE** below 8.
4. **Decide:** $|t| = 1.0 < 1.96$. p-value $= 2 \times \Pr(Z > 1) = 0.317 > 0.05 \Rightarrow$ **do not reject**.
5. **In words:** a gap of 0.3 h is well within normal sampling noise. We **fail to reject** H_0: no evidence against 8 hours (like "not guilty": that does not prove it is exactly 8).

Errors, power, and the role of α



Power $= 1 - \beta$: probability of *not* making a Type II error.

	H_0 true	H_0 false
Reject	Type I (α)	power $1 - \beta$
Keep	correct	Type II (β)

- **Type I** = false alarm; **Type II** = missed detection.
- We **choose** α; β depends on the true μ, n, σ.
- Lower $\alpha \Rightarrow$ cutoff moves right $\Rightarrow \beta$ up.
- The two errors are inversely related: we fix α and try to maximise the power.

Exercise 8: Two-sided test, large sample

A delivery company claims that its **average delivery time is 30 minutes**. You check $n = 225$ random orders and find $\bar{y} = 31.2$ min and $s_{n-1}^2 = 81$.

Question: is the data compatible with the company's claim?

1. Write H_0 and H_A. (The claim could be wrong in either direction, so use a two-sided test.)
2. Compute the standard error $\mathrm{SE}(\bar{y}) = s/\sqrt{n}$.
3. Compute the test statistic $t = \dfrac{\bar{y} - 30}{\mathrm{SE}(\bar{y})}$.
4. Decide: reject H_0 at $\alpha = 5\%$? And at $\alpha = 1\%$?
5. Compute the p-value and check it agrees with step 4.

Hints: $s^2 = 81 \Rightarrow s = 9$. Large n, so use z critical values. p-value $= 2\,(1 - \Phi(|t|))$, where $\Phi(z) = \Pr(Z \le z)$; you may use $\Phi(2.00) = 0.9772$.

Solution · Exercise 8

1. $H_0 : \mu = 30$ vs $H_A : \mu \neq 30$.
2. $\mathrm{SE} = 9/\sqrt{225} = 9/15 = \mathbf{0.6}$ min.
3. $t = \dfrac{31.2 - 30}{0.6} = \mathbf{2.00}$: the sample mean is 2 standard errors above the claim.
4. $\alpha = 5\%$: $|t| = 2.00 > 1.96 \Rightarrow$ **reject** H_0. $\alpha = 1\%$: $2.00 < 2.576 \Rightarrow$ **do not reject**.
5. p-value $= 2\,(1 - 0.9772) = \mathbf{0.0456}$: smaller than 0.05, larger than 0.01. Same decisions as step 4. ✓

In words: at 5% the data are too unlikely if the true mean were 30 minutes, so we doubt the claim. But the evidence is borderline: with a stricter $\alpha = 1\%$ we would not reject.

Exercise 9 · The small-sample trap

A lab sets the target mean strength of a cable at $\mu_0 = 100$. The strength is normally distributed. A sample of $n = 9$ cables gives $\bar{y} = 106.6$ and $s^2_{n-1} = 81$.

1. Compute the t-statistic for $H_0 : \mu = 100$ vs $H_A : \mu \neq 100$.
2. Decide at 5% using (i) the normal critical value and (ii) the correct one. Do they agree?
3. Which is correct and why?

Hint: $n = 9 \Rightarrow$ degrees of freedom $= 8$ ($t_{8;0.975} = 2.306$).

Solution · Exercise 9

1. $\mathrm{SE} = \sqrt{81/9} = 3, \quad t = \dfrac{106.6 - 100}{3} = \mathbf{2.20}.$
2. (i) Normal: $2.20 > 1.96 \Rightarrow$ reject.
   (ii) $t_{8;0.975} = 2.306$: $2.20 < 2.306 \Rightarrow$ **fail to reject**.
3. Only (ii) is valid: $n = 9$ is small and σ is estimated, so the statistic follows t_8. The normal quantile is too small for a small sample and would reject too often.

5. Wrap-up

Summary

One recipe for everything

$$\frac{\text{estimate} - \text{benchmark}}{\text{standard error}} \quad \text{is compared with a } z \text{ or } t \text{ table}$$

- **Confidence interval:** start from the estimate, go $\pm$ (critical value) $\times$ SE. $\Rightarrow$ *plausible values of* μ.
- **Hypothesis test:** measure how many SE the estimate is from the claim μ_0. $\Rightarrow$ *is the claim plausible?*
- **Same ingredients:** $\bar{y}$, $\mathrm{SE} = s/\sqrt{n}$, and the critical value (z for large n or known σ; t_{n-1} for small n with unknown σ).
- **CLT** is what makes all this work for non-normal data when n is large.

Cheat sheet

	Confidence interval	**Test of $H_0 : \mu = \mu_0$**
σ^2 known	$\bar{y} \pm z\,\sigma/\sqrt{n}$	$z = (\bar{y} - \mu_0) \,/\, (\sigma/\sqrt{n})$
σ^2 unknown, large n	$\bar{y} \pm z\,s/\sqrt{n}$	$t = (\bar{y} - \mu_0) \,/\, (s/\sqrt{n}) \approx N(0,1)$
σ^2 unknown, small n, normal Y	$\bar{y} \pm t_{n-1}\,s/\sqrt{n}$	$t = (\bar{y} - \mu_0) \,/\, (s/\sqrt{n}) \sim t_{n-1}$

- $z_{0.95} = 1.645,\ z_{0.975} = 1.96,\ z_{0.995} = 2.576$ · width $= 2\times$ margin
- Reject $\iff p < \alpha \iff |t| > c_{1-\alpha/2}$ (two-sided) · $E \propto 1/\sqrt{n}$